Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
D		2607269

Your Paper's Title(SHUMOJIAYOUZHAN)

Summary

hellow everyone

Keywords: keyword1; keyword2

Contents

1 Introduction

Professional sports are fundamentally an entertainment business, where the core objective is to generate profit and long-term franchise value for owners rather than solely pursuing victories. As stated in the 2026 ICM Problem D, "The player's job is to help his team win" (Cliff Blau), yet owners prioritize that "The player's job is to make money for the owner." Winning enhances fan engagement and revenue, but player popularity, market size, clutch performance, injuries, financing costs, and league structural changes can drive financial outcomes independently of on-court success. The problem highlights this duality through the Women's National Basketball Association (WNBA)'s current transformation—record viewership, franchise expansion, new media deals, and rising player demands—illustrating both growth opportunities and financial risks, though teams from any professional league may be analyzed.

This paper addresses the required tasks for a professional sports team (at least five cooperative players in a league):

- Design a dynamic decision-making model to adjust leverage and operations amid changing performance, economic conditions, and injuries, maximizing profit and value.
- Develop a player acquisition strategy valuing contributions to both competitiveness and profitability, incorporating team dynamics.
- Analyze league expansion impacts, including effects of new team locations.
- Propose one additional business decision (we select dynamic ticket pricing).
- Ensure automatic adaptation to key player injuries.
- Provide a 1–2 page letter to the owner and general manager.

Conventional approaches often treat competitive metrics (e.g., win contributions, clutch and hustle statistics), commercial factors (e.g., brand appeal, market effects), and financial risk management separately. This paper unifies these elements into a single, coherent decision system that is interpretable, practically implementable, and reproducible using publicly available data.

We apply the framework to an NBA team as our primary case study, leveraging its rich historical data for model calibration and backtesting. The approach is general and readily transferable to other leagues, including emerging ones such as the WNBA.

2 Preparation of the Models

2.1 Assumptions

To simplify the complex environment of professional sports and ensure the mathematical tractability of our model, we establish the following core assumptions:

- **Discrete Decision Epochs**: Management evaluates team status and makes strategic adjustments at discrete intervals (e.g., monthly or every 10 games), rather than continuously.
- **Decoupling of Performance and Commercial Value**: A player's total value can be decomposed into independent athletic (OPI) and commercial (CVI) components, which are not always positively correlated.
- **Rational Financial Management**: The team owner acts as a rational economic agent aiming to maximize profit and asset value while strictly adhering to debt constraints such as the Debt Service Coverage Ratio (DSCR).
- **Non-linear Impact of Key Injuries**: The injury of a "key player" has a disproportionate, non-linear downward impact on the team's health index H_t, triggering an immediate defensive shift in management strategy.
- **Price Elasticity of Demand**: Market demand for tickets and merchandise is sensitive to pricing and team performance, following an exponential decay function relative to price increases.
- **Structural Expansion Shocks**: League expansion is treated as an external shock that impacts existing teams through talent dilution, revenue redistribution, and increased travel burdens.

2.2 Notations

To provide a clear structure for our mathematical framework, we categorize the primary symbols into four functional groups: General State, Financial Management, Player Valuation, and Business Strategy.

Table 1: Summary of Notations Categorized by Modeling Tasks

Symbol	Description	Unit/Range
Group 1: General State & Performance (Q1 & Q3)		
t	Decision step or epoch (e.g., monthly or 10-game block)	–
ES_t	Environmental Score (composite signal for state transition)	$[0, 1]$
OPI_t	On-court Performance Index (athletic efficiency metric)	$[0, 1]$
CVI_t	Commercial Value Index (brand asset and monetization potential)	$[0, 1]$
W_t, H_t	Normalized winning percentage and Team Health index	$[0, 1]$
r_t	Prevailing market financing/interest rate index	$[0, 1]$
Group 2: Financial Management & Leverage (Q1)		
L_t	Leverage ratio (Debt-to-Equity ratio)	Ratio
D_{max}	Debt ceiling (maximum sustainable debt based on $DSCR$)	Currency

(Continued on next page)

Symbol	Description	Unit/Range
$EBITDA_t$	Earnings before interest, taxes, depreciation, and amortization	Currency
$DSCR$	Debt Service Coverage Ratio (minimum threshold for solvency)	Ratio
Group 3: Player Valuation & Acquisition (Q2)		
V_i^{sport}	Athletic value of player i	Scalar
V_i^{biz}	Commercial value of player i	Scalar
x_i	Binary decision variable	$\{0, 1\}$
$\lambda(ES)$	State-driven weighting factor	$[0, 1]$
Group 4: Business Strategy - Ticket Pricing		
p_g	Price of a single ticket for game g	Currency
$D_g(p)$	Market demand function for game g	Scalar
A_g	Base demand multiplier	Scalar
k	Price sensitivity coefficient	Constant

3 The Dual-Index Model

Professional sports teams must balance two distinct but interconnected objectives: on-court success and commercial profitability. Winning games drives fan engagement and revenue, yet popularity, market size, and macroeconomic conditions can generate substantial value even in periods of mediocre performance. To capture this duality, we propose a unified decision-making framework centered on two independent yet linkable indices—the On-court Performance Index (OPI) and the Commercial Value Index (CVI)—combined with a dynamic state signal, the Environmental Score (ES), that drives conditional strategy selection through a three-state regime.

3.1 The Dual-Index Framework

3.1.1 On-court Performance Index (OPI)

The On-court Performance Index compresses competitive strength into a single normalized metric in $[0, 1]$:

$$OPI_t = \omega_1 \cdot WinRate_t + \omega_2 \cdot NetRating_t + \omega_3 \cdot PlayoffProb_t,$$

where weights $\omega_1 + \omega_2 + \omega_3 = 1$ (default: $\omega_1 = 0.5$, $\omega_2 = 0.3$, $\omega_3 = 0.2$[1]). When detailed data are unavailable, OPI may be approximated by normalized win rate or Elo rating alone. OPI measures "winning ability" but does not necessarily correlate perfectly with financial returns.

3.1.2 Commercial Value Index (CVI)

The Commercial Value Index quantifies the team's ability to convert performance and brand appeal into revenue and long-term asset value:

$$CVI_t = \nu_1 \cdot Revenue_t + \nu_2 \cdot Engagement_t + \nu_3 \cdot BrandValue_t,$$

with $\nu_1 + \nu_2 + \nu_3 = 1$ (suggested: $\nu_1 = 0.4$, $\nu_2 = 0.3$, $\nu_3 = 0.3$). Revenue includes ticket sales, media rights shares, sponsorships, and merchandise; Engagement encompasses attendance rates and social media interactions; BrandValue is proxied by jersey sales rankings, All-Star votes, and market-adjusted follower growth. CVI explains why teams with high popularity but moderate records can remain financially attractive to owners.

3.2 Environmental Score (ES) as State Signal

Rather than optimizing a single static objective function, we adopt a state-driven conditional optimality approach: strategies adapt based on prevailing conditions. To operationalize this, we define a composite Environmental Score $ES_t \in \mathbb{R}$ that integrates competitive, operational, and macroeconomic signals:

$$ES_t = \alpha W_t + \beta H_t - \gamma r_t, \quad \alpha + \beta + \gamma = 1,$$

where

- $W_t \in [0, 1]$: normalized recent win rate,
- $H_t \in [0, 1]$: health/availability index (weighted attendance rate of core rotation players),
- $r_t \in [0, 1]$: normalized financing cost (e.g., federal funds rate or 10-year Treasury yield proxy).

Default weights are $\alpha = 0.4$, $\beta = 0.3$, $\gamma = 0.3$. High win rates and player health improve investment attractiveness, while rising interest rates increase the cost of leverage.

3.3 Three-State Regime and Action Mapping

The value of ES_t determines the operating state and corresponding strategic posture:

ES Range	State	Target Leverage
$ES_t > 0.6$	Aggressive	High (≈ 1.0)
$0.3 \le ES_t \le 0.6$	Neutral	Moderate (≈ 0.5)
$ES_t < 0.3$	Defensive	Low (≈ 0)

Table 2: Three-state regime driven by Environmental Score.

- **Aggressive State ($ES_t > 0.6$):** Favor growth and championship pursuit. Target high leverage to finance star acquisitions, premium ticket pricing, and aggressive marketing.

- **Neutral State ($0.3 \leq ES_t \leq 0.6$):** Balance growth and stability. Maintain moderate leverage, pursue cost-effective roster improvements, and optimize operations.
- **Defensive State ($ES_t < 0.3$):** Prioritize cash-flow safety and risk reduction. Delever aggressively, shed high-risk contracts, and focus on attendance retention through stable pricing.

This state machine enables conditional optimality: in favorable environments the framework favors growth and championship pursuit; in adverse conditions it prioritizes cash-flow safety and risk reduction. Safety constraints (Debt Ceiling and Equity Trigger, detailed in Section 4) prevent excessive risk even in Aggressive states.

The unified Dual-Index Model with ES-driven states serves as the core engine for all subsequent analyses: dynamic leverage control (Q1), player acquisition weighting (Q2), expansion impact assessment (Q3), ticket pricing policy, and automatic injury response.

4 Model I: Dynamic Leverage Control Model

5 Model II: Player Acquisition Strategy

To develop a player acquisition strategy for the upcoming season, we extend the Dual-Index Model (Section 3) by evaluating players through a dual-value lens that explicitly accounts for both on-court contributions and commercial profitability. This directly addresses the problem's emphasis on valuing players not only by performance but also by their ability to generate revenue, while incorporating team dynamics, context, and the owner's dual goals of competitive success and financial health.

Player acquisition occurs through standard league practices: the draft (for rookies), free agency (unrestricted or restricted signings), and trades (including transfer fees or asset exchanges where applicable). Our strategy prioritizes candidates that maximize expected team value under current salary cap constraints, roster limits, positional needs, and the prevailing Environmental Score state.

5.1 Player Valuation: Dual-Value Framework with Team Context

Each player i is assigned a composite value V_i that combines competitive and commercial dimensions, adjusted for team-specific synergies and risks.

5.1.1 Competitive Value V_i^{sport}

The competitive contribution is modeled as:

$$V_i^{sport} = Base_i + a \cdot Clutch_i + b \cdot Hustle_i - c \cdot Risk_i + d \cdot Potential_i - e \cdot SynergyPenalty_i$$

where

- $Base_i$: Baseline performance (e.g., normalized Win Shares, PER, or BPM).
- $Clutch_i$: Clutch performance in high-leverage moments (e.g., net efficiency in last 5 minutes of close games).
- $Hustle_i$: Effort-based contributions (e.g., deflections, screens, contested rebounds).
- $Risk_i$: Injury risk (e.g., 1 minus average games played over recent seasons).
- $Potential_i$: Upside proxy (e.g., age-adjusted improvement curve; higher for younger players).
- $SynergyPenalty_i$: Contextual penalty for diminishing marginal returns and negative synergies, using usage rate (USG%) as a proxy for ball-dominant playstyles:

  $$SynergyPenalty_i = f \cdot \max(0, USG_i - TeamAvgUSG)^2$$

  Here, USG_i is the player's recent usage rate, $TeamAvgUSG$ is the weighted average usage of the current roster (updated iteratively during optimization), and $f > 0$ is a calibrated coefficient (suggested range 0.4–0.8). The quadratic form heavily penalizes excessive overlap in high-usage players, reflecting real-world efficiency drops when multiple ball-dominant stars compete for touches. Low-usage role players receive no penalty and may implicitly benefit in complementary lineups.

Default coefficients (a, b, c, d, e, f) are tuned via historical data; sensitivity analysis (Section **??**) confirms robustness.

5.1.2 Commercial Value V_i^{biz}

Commercial impact captures off-court revenue generation:

$$V_i^{biz} = Brand_i \times MarketMultiplier$$

where $Brand_i$ aggregates normalized metrics such as social media followers, jersey sales rank, and All-Star voting totals, and $MarketMultiplier$ reflects the team's market size (e g., higher for large-market franchises due to amplified sponsorship and media opportunities).

5.1.3 Composite Value with State-Dependent Weighting

The total value is a dynamic convex combination driven by the current ES state:

$$V_i = \lambda(ES_t) \cdot V_i^{sport} + (1 - \lambda(ES_t)) \cdot V_i^{biz}$$

Recommended λ values:

- Aggressive state ($ES_t > 0.6$): $\lambda \approx 0.7$ (prioritize immediate on-court impact for championship contention).
- Neutral state ($0.3 \leq ES_t \leq 0.6$): $\lambda \approx 0.5$ (balanced approach).
- Defensive state ($ES_t < 0.3$): $\lambda \approx 0.3$ (favor low-risk, high-brand players to preserve cash flow and flexibility).

This weighting ensures acquisition aligns with the team's immediate needs as signaled by the broader decision framework.

5.2 Acquisition Optimization and Constraints

We formulate acquisition as a constrained optimization problem to select or retain a roster that maximizes net value:

$$\max \sum_i x_i \left(V_i - \eta \cdot Cost_i\right)$$

where $x_i \in \{0, 1\}$ indicates selection/retention of player i, $Cost_i$ includes salary, luxury-tax impact, and trade assets, and $\eta > 0$ is a cost-aversion parameter.

Key constraints include:

- Salary cap and luxury-tax thresholds: $\sum x_i \cdot Salary_i \leq Cap + TaxThreshold$.
- Roster size: exactly or at most the league-mandated number of players.
- Positional balance: minimum requirements per position/role (e.g., guards, forwards, centers).
- Risk budget: $\sum x_i \cdot Risk_i \leq RiskBudget$ to limit exposure to injuries.
- Synergy update: $TeamAvgUSG$ recalculated iteratively after provisional selections.

The problem can be solved heuristically (greedy ranking by value-per-cost with positional fixes) or exactly via integer linear programming for small instances. In practice, managers iterate candidates from draft prospects, free agents, and trade targets.

5.3 Strengths and Limitations

This strategy excels at reconciling performance and profitability, dynamically adapting to team state, and explicitly modeling contextual fit via the synergy penalty—avoiding common pitfalls of overpaying for individually talented but poorly fitting stars. Limitations include reliance on accurate Brand and USG data (which can be noisy) and the need for periodic coefficient recalibration using historical outcomes (addressed in backtesting, Section 6).

6 Sensitivity Analysis And Backtesting

7 Model Evaluation and Further Discussion

7.1 Strengths

- **Applies widely**
  This system can be used for many types of airplanes, and it also solves the interference during the procedure of the boarding airplane,as described above we can get to the optimization boarding time.We also know that all the service is automate.
- **Improve the quality of the airport service**
  Balancing the cost of the cost and the benefit, it will bring in more convenient for airport and passengers.It also saves many human resources for the airline.
-

8 Conclusion

Suspendisse vel felis. Ut lorem lorem, interdum eu, tincidunt sit amet, laoreet vitae, arcu. Aenean faucibus pede eu ante. Praesent enim elit, rutrum at, molestie non, nonummy vel, nisl. Ut lectus eros, malesuada sit amet, fermentum eu, sodales cursus, magna. Donec eu purus. Quisque vehicula, urna sed ultricies auctor, pede lorem egestas dui, et convallis elit erat sed nulla. Donec luctus. Curabitur et nunc. Aliquam dolor odio, commodo pretium, ultricies non, pharetra in, velit. Integer arcu est, nonummy in, fermentum faucibus, egestas vel, odio.

References

[1] D. E. KNUTH The TEXbook the American Mathematical Society and Addison-Wesley Publishing Company , 1984-1986.

[2] Lamport, Leslie, LATEX: " A Document Preparation System ", Addison-Wesley Publishing Company, 1986.

[3] http://www.latexstudio.net/

[4] http://www.chinatex.org/

Appendices

Appendix A First appendix

Aliquam lectus. Vivamus leo. Quisque ornare tellus ullamcorper nulla. Mauris porttitor pharetra tortor. Sed fringilla justo sed mauris. Mauris tellus. Sed non leo. Nullam elementum, magna in cursus sodales, augue est scelerisque sapien, venenatis congue nulla arcu et pede. Ut suscipit enim vel sapien. Donec congue. Maecenas urna mi, suscipit in, placerat ut, vestibulum ut, massa. Fusce ultrices nulla et nisl. Here are simulation programmes we used in our model as follow.

Input matlab source:

```
function [t,seat,aisle]=OI6Sim(n,target,seated)
pab=rand(1,n);
for i=1:n
    if pab(i)<0.4
       aisleTime(i)=0;
    else
        aisleTime(i)=trirnd(3.2,7.1,38.7);
    end
end
```

Appendix B Second appendix

some more text **Input C++ source:**

```
//===========================================================================
// Name        : Sudoku.cpp
// Author      : wzlf11
// Version     : a.0
// Copyright   : Your copyright notice
// Description : Sudoku in C++.
//===========================================================================

#include <iostream>
#include <cstdlib>
#include <ctime>

using namespace std;

int table[9][9];

int main() {

    for(int i = 0; i < 9; i++){
        table[0][i] = i + 1;
    }

    srand((unsigned int)time(NULL));

    shuffle((int *)&table[0], 9);

    while(!put_line(1))
    {
        shuffle((int *)&table[0], 9);
    }

    for(int x = 0; x < 9; x++){
```

```
        for(int y = 0; y < 9; y++){
            cout << table[x][y] << " ";
        }

        cout << endl;
    }

    return 0;
}
```

References

[1] FiveThirtyEight. (2025) How our nba predictions work. Methodology article, updated continuously. Core modeling framework described in post-2020 versions. [Online]. Available: https://fivethirtyeight.com/methodology/how-our-nba-predictions-work/